RENVOLT

PRELIMINARY ENGINEERING REPORT

Phase 1: Battery-Backed DC Fast Charging Infrastructure

Prepared For: ZEVIP Grant Application & Phase 1 Site Hosts

Prepared By: Renvolt Engineering

Date: March 14, 2026

Status: CONFIDENTIAL DRAFT

Standard: Canadian Electrical Code (CEC) CSA C22.1-2024 | CSA C22.2 No. 107.1
Jurisdiction: Alberta, Canada — FortisAlberta / ATCO Electric service territory
Architecture: 150 kW Shared DC Bus / 100 kWh Local BESS Storage

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1. Design Philosophy

This system is designed as a **zero-export, behind-the-meter** installation. The battery charges slowly from the grid (≤50 kW) and discharges rapidly (150 kW) into EV dispensers. From the utility's perspective, this site is a **predictable 50–70 kW commercial load** — not a generator.

> *[!IMPORTANT] Zero-export configuration avoids the Generator Interconnection Application (GIA) requirement with FortisAlberta/ENMAX. The system qualifies as standard commercial service under AUC Rule 007 Section 3.*

Operating Modes

Mode	Grid Draw	Battery State	DCFC Output
Idle (no vehicles)	30–50 kW	Charging	0 kW
Single vehicle	50 kW	Discharging	150 kW (ZEVIP Compliant)
Multi-vehicle (2 stalls)	50 kW	Discharging	75 kW × 2 (Dynamic Share)
Battery depleted	50 kW (passthrough)	Depleted — recharging	50 kW (degraded mode)
Off-peak recovery	50 kW	Charging from 0–100%	0 kW

Key metric: Full battery recovery from 0% to 100% = 100 kWh ÷ 50 kW = **2 hours**

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2. System Architecture

```
graph TD
    subgraph UTILITY["Utility Service (FortisAlberta)"]
        GRID["Grid<br/>208Y/120V 3ϕ 4W<br/>200A Service"]
    end

    subgraph MAIN_PANEL["Main Distribution Panel"]
        MB["Main Breaker<br/>200A / 3P"]
        M1["Meter<br/>CT-rated Revenue Meter"]
    end

    subgraph BATTERY_SYSTEM["Battery Energy Storage System (BESS)"]
        INV["Bidirectional Inverter<br/>50 kW AC-DC<br/>208V AC → 800V DC"]
        BMS["Battery Management System<br/>Cell balancing / Thermal / SOC"]
        BATT["LFP Battery Bank<br/>100 kWh @ 819V nominal<br/>16 × 51.2V modules in series"]
        DC_DISC["DC Disconnect<br/>250A / 1000V DC"]
    end

    subgraph DCFC["DC Fast Charging Output (Dynamic Share)"]
        DCDC["Central Power Cabinet<br/>800V → 200-920V<br/>150 kW Shared DC Bus"]
```

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    D1["Dispenser Post 1<br/>CCS1 + NACS<br/>Up to 150 kW"]
    D2["Dispenser Post 2<br/>CCS1 + NACS<br/>Up to 150 kW"]
end

subgraph PROTECTION["Protection & Controls"]
    GFP["Ground Fault<br/>Protection"]
    SPD["Surge Protection<br/>Device"]
    ATS["Transfer Logic<br/>(Zero-Export Controller)"]
    FIRE["Fire Suppression<br/>LFP-rated"]
end

GRID --> M1 --> MB
MB --> INV
INV <--> BATT
BMS --> BATT
DC_DISC --> BATT
BATT --> DCDC
DCDC --> D1
DCDC --> D2
GFP --> MB
SPD --> MB
ATS --> INV
FIRE --> BATT
```

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3. Electrical Service Requirements

3.1 Utility Service Sizing

The grid connection only needs to support the **battery charge rate**, not the full 150 kW output. This is the core advantage.

Parameter	Without Battery	With Battery
Peak grid demand	**150 kW**	**50 kW**
Service size needed	600A @ 208V	**200A @ 208V**
Estimated service upgrade cost	$80,000–$150,000	**$0–$15,000**
Monthly demand charge (@ $12/kW)	$1,800/mo	**$600/mo**

3.2 Service Specification

Item	Specification
Voltage	208Y/120V, 3-phase, 4-wire
Service size	200A
Service entrance	Overhead or underground per utility
Meter	CT-rated revenue meter (utility-supplied)

Service conductor	3/0 AWG copper, TECK90, per CEC Table 2
Grounding electrode	Per CEC Rule 10-700 (min. #6 AWG copper to ground rod)
Bonding	Per CEC Rule 10-600 series

3.3 Alternative: 600V Service

For sites with existing 347/600V 3-phase service (common in Alberta industrial areas):

Item	Specification
Voltage	347/600V, 3-phase, 4-wire
Service size	100A (50 kW at 600V = ~48A, 125% = 60A)
Service conductor	#4 AWG copper, TECK90
Advantage	Lower current = smaller wire, lower losses
Disadvantage	Requires step-down transformer if host building uses 208V

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4. Wire Sizing Calculations

All calculations per **CEC 2024**, using **75°C copper conductors** and **CEC Table 2** ampacities.

4.1 AC Side: Grid → Bidirectional Inverter

Load: 50 kW continuous at 208V, 3-phase

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Current (I) = P / (V × √3 × PF)
           = 50,000 / (208 × 1.732 × 0.95)
           = 50,000 / 342.1
           = 146.1 A

CEC Rule 8-104(5): Continuous load × 125%
           = 146.1 × 1.25
           = 182.6 A → Use 200A breaker
```

Component	Specification	CEC Reference
Conductors	3/0 AWG copper, TECK90 (200A @ 75°C)	Table 2
Neutral	3/0 AWG copper (full-sized)	Rule 4-022
Ground	#6 AWG copper (bonding conductor)	Table 16
Conduit	2" rigid metal conduit (RMC) or 2" TECK cable	Table D16
Breaker	200A, 3-pole, 42 kAIC minimum	Rule 14-012
Run length	≤ 25 m (voltage drop < 3%)	Rule 8-102

Voltage Drop Check (25 m run):

```
Vd = (2 × L × I × R) / 1000
   = (2 × 25 × 146 × 0.328) / 1000     [R for 3/0 AWG = 0.328 Ω/km]
   = 2.39 V per conductor
   = 2.39 / 208 × 100 = 1.15%           ✓ Under 3% limit
```

4.2 DC Side: Battery Bank → DC-DC Converter → Dispensers

Load: 150 kW at 800V DC bus

```
Current (I) = P / V
           = 150,000 / 800
           = 187.5 A

Continuous rating × 125%:
           = 187.5 × 1.25
           = 234.4 A → Use 250A DC fuse/breaker
```

Component	Specification	Notes
DC bus conductors	4/0 AWG copper, PV Wire (USE-2), 600V DC min	230A @ 75°C per Table 2
DC disconnect	250A, 1000V DC rated, lockable	Required per CEC Rule 64-112
DC fuses	250A, 1000V DC, gPV type	Per CEC Rule 64-060
Conduit	1.5" rigid metal or PV-rated tray cable	

4.3 DC Side: Power Cabinet → Individual Dispenser Posts

Load per dispenser: Dynamic. Up to 150 kW (ZEVIP Compliant) for a single vehicle, or mathematically shared across two vehicles.

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Worst case (single vehicle demanding full 150kW at 400V):
I = 150,000 / 400 = 375 A (Requires liquid-cooled dispenser cable or 400A continuous rating)

Shared case (two vehicles demanding 75kW at 400V):
I = 75,000 / 400 = 187.5 A → 125% = 234 A
```

Component	Specification
Conductors (per dispenser)	500 kcmil copper or specialized liquid-cooled assembly
Breaker per dispenser	400A-500A, 1000V DC
Cable length	≤ 10 m (dispenser cord is integral to unit)

4.4 Wire Schedule Summary

Circuit	Voltage	Current	Wire	Breaker/Fuse	Conduit
Grid → Main panel	208V 3φ AC	146A (182A rated)	3/0 AWG Cu TECK90 × 3 + N + G	200A 3P	2" RMC

Main panel → Inverter	208V 3φ AC	146A	3/0 AWG Cu TECK90 × 3 + N + G	200A 3P	2" RMC
Inverter ↔ Battery	800V DC	62.5A (charge) / 187.5A (discharge)	4/0 AWG Cu USE-2 × 2 + G	250A DC	1.5" RMC
Battery → DC-DC converter	800V DC	187.5A	4/0 AWG Cu USE-2 × 2 + G	250A DC	1.5" RMC
Cabinet → Dispenser 1	200–920V DC	up to 375A	500 kcmil / Liquid-Cooled	400A DC	Integral
Cabinet → Dispenser 2	200–920V DC	up to 375A	500 kcmil / Liquid-Cooled	400A DC	Integral
Host building sub-panel	208V 3φ AC	40A	#8 AWG Cu TECK90 × 3 + N + G	50A 3P	1" RMC

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5. Battery Bank Specification

5.1 Cell Chemistry & Configuration

Parameter	Specification
Chemistry	LFP (Lithium Iron Phosphate — $LiFePO_4$)
Why LFP	No thermal runaway, 5000+ cycle life, –30°C operation, no cobalt
Total capacity	100 kWh usable (110 kWh gross at 90% DoD)
Nominal voltage	819.2V (16 modules × 51.2V/module)
Module configuration	16S rack module, 51.2V, 100 Ah per module
Modules in series	16 (total: 819.2V nominal)
Min voltage (empty)	704V (16 × 44V cutoff)
Max voltage (full)	921.6V (16 × 57.6V float)

5.2 Charge / Discharge Rates

Parameter	Value	C-Rate
Max charge rate	50 kW (from grid)	0.5C
Max discharge rate	150 kW (to DCFC)	1.5C
Full charge time	2 hours (0–100%)	—
Full discharge time	40 minutes (at 150 kW)	—
Cycle life	>5,000 cycles @ 80% DoD	—

Calendar life	15+ years	—

5.3 Thermal Management

Condition	System Response
>45°C cell temp	Reduce charge/discharge to 50%
>55°C cell temp	Shutdown — thermal fault
<–20°C cell temp	Activate heated enclosure, delay charge until >0°C
<–40°C ambient	Enclosure heaters maintain battery at >5°C (500W heating element)
Normal range	15–35°C (liquid or forced-air cooled)

5.4 BMS Requirements

- Cell-level voltage monitoring (±10 mV accuracy)
- Cell-level temperature monitoring (NTC per module)
- Active cell balancing (top-balance at 3.65V/cell)
- SOC estimation (Coulomb counting + OCV correction)
- Communication: CAN bus to inverter, Modbus TCP to site controller
- Safety: pre-charge circuit, contactor control, fuse monitoring

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6. Single-Line Diagram

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┌──────────────────────────────────────────────────────────────────────────┐
│  UTILITY (FortisAlberta)                                                 │
│  208Y/120V, 3ϕ, 4W                                                       │
└──────────────────────────────────────────────────────────────────────────┘
                              │
                              │
                          ┌───┴───┐
                          │ CT Meter │  (Revenue meter, utility-owned)
                          └───┬───┘
                              │  3/0 AWG Cu TECK90 × 4 + #6G in 2" RMC
                              │
                     ┌────────┴──────────┐
                     │   MAIN BREAKER    │
                     │   200A / 3P       │
                     │   42 kAIC min     │
                     └──────┬────┬───────┘
                            │    │
          ┌─────────────────┘    └─────────────────┐
          │                                        │
      ┌───┴────┐                              ┌────┴─────┐
      │ SPD    │                              │ Host Sub-panel│
      │ Type 2 │                              │ 50A / 3P      │
      └────────┘                              │ #8 AWG Cu     │
          │                                   │ Lights, CCTV, │
          │                                   │ Signage, HVAC │
```

```
    ┌────┴──────────────┐                    └────────────────┘
    │ BIDIRECTIONAL     │
    │ INVERTER          │
    │ 50 kW AC ↔ DC     │
    │ 208V AC → 800V DC │
    │ 97.5% efficiency  │
    └───────────────────┘
             │
             │  4/0 AWG Cu USE-2 × 2 + G in 1.5" RMC
             │
     ┌───────┴─────────┐
     │ DC DISCONNECT   │
     │ 250A / 1000V    │
     │ Lockable        │
     └─────────────────┘
             │
             │
     ┌───────┴─────────┐         ┌──────────────────┐
     │ LFP BATTERY     │◄────────►│ BMS CONTROLLER   │
     │ 100 kWh         │  CAN    │ Cell monitoring   │
     │ 819.2V nom      │  bus    │ Thermal mgmt      │
     │ 16 modules      │         │ SOC estimation     │
     │ (16S × 51.2V)   │         │ Contactor control  │
     └─────────────────┘         └──────────────────┘
             │
             │  4/0 AWG Cu USE-2 × 2 + G
             │
     ┌───────┴─────────┐
     │ DC FUSE         │
     │ 250A / 1000V    │
     │ gPV type        │
     └─────────────────┘
             │
             │
    ┌────────┴──────────────┐
    │ POWER CABINET         │
    │ 800V → 200-920V       │
    │ 150 kW total          │
    │ Dynamic power share   │
    └──────┬────────────────┬┘
           │                │
   ┌───────┴────┐   ┌───────┴────┐
   │DISP. 1     │   │DISP. 2     │
   │Up to       │   │Up to       │
   │150 kW      │   │150 kW      │
   │CCS1+NACS   │   │CCS1+NACS   │
   └────────────┘   └────────────┘


GROUNDING SYSTEM
═══════════════
Equipment ground: #6 AWG Cu to main panel ground bus
Grounding electrode: 2× 10' ground rods, min 3m apart
Bond: #6 AWG Cu between rods
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Battery enclosure: bonded to ground bus
DCFC dispensers: equipment ground per CEC Rule 10-400
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7. Protection & Safety

7.1 AC Protection

Device	Rating	Purpose	CEC Rule
Main breaker	200A, 3P, 42 kAIC	Overcurrent, service disconnect	14-012
SPD (Type 2)	208V, 100 kA	Surge protection	18-100
Ground fault (GFCI)	30 mA, Class A	Personnel protection at dispensers	26-700
AFCI	Per panel schedule	Arc fault (if required by AHJ)	26-656

7.2 DC Protection

Device	Rating	Purpose
DC disconnect	250A, 1000V DC, lockable	Battery isolation (maintenance/fire)
DC fuses (gPV)	250A, 1000V DC	Short circuit protection
Pre-charge relay	10A, 1000V DC	Inrush current limiting at startup
Contactor (×2)	250A, 1000V DC	BMS-controlled battery isolation

7.3 Fire Safety

Requirement	Specification
Enclosure rating	NEMA 3R (outdoor, rain-tight) minimum
Fire suppression	Clean agent (Novec 1230 or aerosol) — NOT water or foam
Thermal runaway detection	Gas sensor (VOC) + rate-of-rise thermal sensor
Emergency stop	Exterior-mounted E-stop, red mushroom, opens DC contactors
Clearances	1 m from building walls, 3 m from combustibles (per NFPA 855)
Signage	"DANGER: BATTERY ENERGY STORAGE — 800V DC" at all access points
First responder placard	Per NFPA 855 Section 4.1.5 — posted at site entrance

7.4 Arc Flash Labeling

Location	Estimated Incident Energy	PPE Category
Main panel (AC)	4–8 cal/cm^2	Cat 1
DC disconnect	2–4 cal/cm^2 (limited by fuse)	Cat 1

Battery rack	8–15 cal/cm²	Cat 2

> *[!WARNING] Arc flash study **must** be performed by a P.Eng. or certified technician before energization. These are estimates only.*

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8. Site Layout Considerations

8.1 Equipment Footprint

Equipment	Dimensions (approx.)	Weight
Battery enclosure (outdoor rated)	2.4 m × 1.2 m × 2.1 m (8' × 4' × 7')	~2,500 kg
Inverter + DC-DC (integrated unit)	1.5 m × 0.8 m × 2.0 m	~800 kg
Main distribution panel	0.9 m × 0.3 m × 1.5 m	~100 kg
DCFC Dispenser (each)	0.6 m × 0.5 m × 1.8 m	~300 kg
Total pad area	**~3.5 m × 6 m (120 sq ft)** concrete pad	

8.2 Pull-Through Bay Layout

```
                ┌──────────────────────────────┐
                │       EQUIPMENT PAD          │
                │  [BATT]  [INV/DCDC]  [PANEL]  │
                │       (fenced, 1.8m chain-link)  │
                └──────────────┬───────────────┘
                               │  underground conduit
   ────────────────────────────┼──────────────────────────
 ← ENTRY                       │                EXIT →
   ─────┌──────────────[DISP 1]──────────────┐─────
        │           Bay 1 (pull-through)     │
   ─────└────────────────────────────────────┘─────
   ─────┌──────────────[DISP 2]──────────────┐─────
        │           Bay 2 (pull-through)     │
   ─────└────────────────────────────────────┘─────
   ──────────────────────────────────────────────────
                 HIGHWAY →
```

Pull-through clearance: 25 m length × 4 m width per bay (accommodates truck + 16 m trailer)

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9. Bill of Materials (Preliminary)

Item	Qty	Unit Cost (CAD)	Total (CAD)
LFP Battery modules (51.2V/100Ah)	16	$3,500	$56,000
Battery rack + enclosure (NEMA 3R, heated)	1	$12,000	$12,000
BMS controller + harness	1	$5,000	$5,000

Bidirectional inverter (50 kW, 208V↔800V)	1	$18,000	$18,000
Central Power Cabinet (150kW shared DC bus)	1	$22,000	$22,000
DCFC Dispenser Post (Capable of 150kW each)	2	$15,000	$30,000
Main distribution panel (200A)	1	$3,500	$3,500
DC disconnect (250A/1000V)	2	$1,200	$2,400
SPD, fuses, breakers, contactors	1 lot	$4,000	$4,000
Cable & conduit (AC + DC)	1 lot	$6,000	$6,000
Concrete pad + bollards + fencing	1	$15,000	$15,000
Fire suppression (Novec 1230)	1	$5,000	$5,000
Signage, markings, lighting	1 lot	$3,000	$3,000
Electrical installation labour	~120 hrs	$120/hr	**$14,400**
Engineering (P.Eng. stamp, drawings)	1	$8,000	$8,000
Permitting + inspection fees	1	$3,000	$3,000
TOTAL (pre-grant)			**$207,300**
ZEVIP 75% (Indigenous partnership)			**–$155,475**
NET COST TO RBS			**$51,825**

> *[!NOTE] These are Q1 2026 estimates. Battery module prices are declining ~15%/year. Actual costs will vary by site conditions, utility requirements, and equipment vendor.*

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10. Permit & Inspection Checklist

Step	Responsible	Timeline
Electrical permit application (Alberta Safety Codes)	Master Electrician	Week 1
FortisAlberta service application (200A commercial)	RBS / Electrician	Weeks 1–4
P.Eng. stamped drawings (single-line, site plan)	Contracted P.Eng.	Weeks 2–3
AHJ review (municipal planning, if applicable)	Municipality	Weeks 3–6
Foundation / concrete pour	Civil contractor	Week 5
Equipment installation	RBS Master Electrician	Weeks 6–8
Rough-in inspection	Safety Codes Officer	Week 7
Energization request to FortisAlberta	RBS / Utility	Week 8
Final inspection + system commissioning	Safety Codes Officer + RBS	Week 9

CUL/CSA compliance verification	Equipment vendor	Pre-shipment
Total timeline: ~9–12 weeks (from permit to energized)		

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11. Key Standards & References

Standard	Application
CEC CSA C22.1-2024	All electrical installation requirements
CSA C22.2 No. 107.1	Power conversion equipment
CSA C22.2 No. 340	Battery ESS — safety requirements
UL 9540 / UL 9540A	ESS installation + thermal runaway testing
NFPA 855	Stationary energy storage systems
SAE J1772 / J3068	AC charging connector standards
SAE J3400	NACS connector standard
IEC 61851-23	DC EV supply equipment
AUC Rule 007	Alberta generator/storage interconnection

> *[!CAUTION] This document is for **preliminary design and planning purposes only.** All electrical designs must be reviewed and stamped by a licensed Professional Engineer (P.Eng.) registered in Alberta before construction. Wire sizing, protection coordination, and arc flash calculations must be verified against site-specific conditions.*